

Input

`p = 23`

`g = 5`

`a = 6 # Alice's private key`

`b = 15 # Bob's private key`

`A = pow(g, a, p)`

`B = pow(g, b, p)`

`K_alice = pow(B, a, p)`

`K_bob = pow(A, b, p)`

`print("Alice sends:", A)`

`print("Bob sends:", B)`

`print("Shared Secret Key (Alice):", K_alice)`

`print("Shared Secret Key (Bob):", K_bob)`

Output

`Alice sends: 8`

`Bob sends: 19`

`Shared Secret Key (Alice): 2`

`Shared Secret Key (Bob): 2`